

MATRIX SYSTEMS

5591 LEO PARK RD
WEST JORDAN, UT 84081
(801) 930-9492



Project Number:	Matrix		
Project Name:	Matrix Disconnect box		
Drawing Number:	MTX-xxxx		
Product ID:	PLC_001		
Product Description:	Buyoff Disconnect box		
Customer:	Matrix		
Customer Job No.:	XXXXX		
Commission:			
Manufacturing Date:	09-29-2025		
Power Supply	480 Vac / 3Φ / 60 Hz	Env. Protection Rating	Type 12
Main Breaker Rating	25 A	Largest Motor	
SCCR	14 kA	Static Heat Load	
Min Feeder Size	14 AWG	Duty Heat Load	
Control Voltage	24 VDC / 120 VAC	Weight	
Created	2026-05-18	Andrew Brunt	
Last Modified	2026-05-18	Andrew Brunt	

REV	REVISION COMMENT	DATE	DESIGNER
01	Initial release	2026-05-18	Andrew Brunt

☐ APPROVED	APPROVED BY (PRINT NAME)	POSITION
☐ APPROVED W/ REVISIONS NOTED		
☐ REJECTED, RESUBMIT REQUIRED	SIGNATURE	DATE
NOTE: APPROVAL SIGNATURE IS REQUIRED BEFORE PROCUREMENT, FABRICATION OR PRODUCTION WILL BEGIN. ANY COMMERCIAL IMPACT IN THE FORM OF CHANGES, REVISIONS, OR MODIFICATIONS MADE AFTER PRODUCTION STARTS MAY RESULT IN DELAYS OR ADDITIONAL COSTS.		

NOTES:

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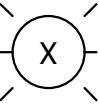


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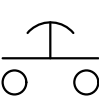
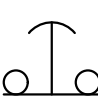
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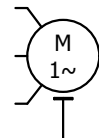
OUTPUT CONTROL SYMBOLS

SYMBOL	DESCRIPTION
	RELAY COIL (CR) CONTACTOR (M)
	SOLENOID (SOL)
	PILOT LIGHT X = COLOR (LT) or (PBL)
	ALARM HORN (AH)
	HEATING ELEMENT (HTR)

INPUT CONTROL SYMBOLS

NORMALLY OPEN	NORMALLY CLOSED	DESCRIPTION
		PUSHBUTTON (PB) or (PBL)
		SELECTOR SWITCH (SS) or (SSL)
		MUSHROOM PUSHBUTTON (PB) or (PBL)
		LIMIT SWITCH (LS)
		AUXILIARY CONTACT

POWER DIST. SYMBOLS

SYMBOL	DESCRIPTION
	DISCONNECT SWITCH (DS)
	CIRCUIT BREAKER (CB)
	FUSE (FU)
	MOTOR OVERLOAD (OL)
	MOTOR STARTER (M)
	3-PHASE MOTOR (MTR)

WIRING COLORS

Ungrounded Power Circuits	Black	(BLK)
Grounded AC Power Circuits	White	(WHT)
Protective wire	Green	(GRN)
Ungrounded AC Control Circuits	Red	(RED)
Grounded AC Control Circuits	White	(WHT)
Ungrounded DC Control Circuits	Blue	(BLU)
Grounded DC Control Circuits	White/Blue	(W/B)
External Voltage Source	Orange	(ORG)
Excepted Circuits	Orange	(ORG)

MIN WIRING SIZES

CIRCUIT	SIZE	MAX AMPACITY
Power Circuits	14 AWG	15 A
Control Circuits	16 AWG	10 A
	18 AWG	7 A
PLC I/O	20 AWG	5 A
	22 AWG	3 A
	24 AWG	2 A
NOTE: Wire sizes must comply with UL 508A sections 29.6 and 38.2. Use Table 28.1 to determine correct wire sizes for loads that exceed the max ampacity listed here. For circuits that are less than 100A, use the 60° F column. For circuits greater than 100A, use the 75° F column.		

Symbols in accordance with UL 508A

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Bill of Materials: MTX-xxxx (PLC_001)

Idx	Qty	Part Number	Part Description	Manufacturer	Supplier	Customer Supplied	Device Tag	Location	Remarks
1	1 ea	R-3W	VOLTAGE INDICATOR 40-750VAC, 3-Phase, Type 4X, 12, 13, 6' CABLE.	Grace Technologies	COD	☒	LT105	MTX-xxxx	
2	2 ea	09300160301	Bulkhead mounted housing, Size: 16 B, Low construction, Double locking lever, Polycarbonate (PC), Stainless steel	HARTING		☒	PLUG_INCOMING_FEMALE;PLUG_OUTGOING_FEMALE	MTX-xxxx	
3	3 ea	09310062601	Inserts, Screw termination, Male, Polycarbonate (PC), RAL 7032 (pebble grey), Rated current: 35 A, Size: 16 B, Contacts: 6, Conductor cross-section: 1.5 ... 6 mm², Copper alloy, Silver plated	HARTING		☒	U145;U155;U155.1	MTX-xxxx	
4	2 ea	09310062701	Inserts, Screw termination, Female, Polycarbonate (PC), RAL 7032 (pebble grey), Rated current: 35 A, Size: 16 B, Contacts: 6, Conductor cross-section: 1.5 ... 6 mm², Copper alloy, Silver plated	HARTING		☒	PLUG_INCOMING_FEMALE;PLUG_OUTGOING_FEMALE	MTX-xxxx	
5	3 ea	19000005098	Cable gland, M40, Clamping range: 22 ... 32 mm, Material (cable glands): Brass, Material (clamping ring): Polyamide (PA)	HARTING		☒	U145;U155;U155.1	MTX-xxxx	
6	3 ea	19300160428	Hood, Size: 16 B, High construction, Double locking lever, Top entry, 1x M40	HARTING		☒	U145;U155;U155.1	MTX-xxxx	
7	1 ea	CP1616	Concept Panel, fits 16.00x16.00 inch Enclosure, White, Mild Steel	Nvent Hoffman	HOF	☒	177	MTX-xxxx	
8	1 ea	CSD161610	Concept Single-Door Enclosure, 16.00x16.00x10.00, Gray, Steel	Nvent Hoffman	HOF	☒	174	MTX-xxxx	
9	1 ea	2471N5	Static-Control Unthreaded-Hole Pull Handle, Black Nylon, 7-1/16" Center-to-Center Mount Hole Width	McMaster Carr	MCM	☒	5	MTX-xxxx	
10	4 ft	Duct, 1x3 , Gray	Wiring Duct Narrow Slot, 1 Inches x 3 Inches, Gray PVC	Matrix Systems	RWE	☒	178...181	MTX-xxxx	
11	2 ea	TP-1X4	Panel Tag Plate, 1x4 in, Custom Text	Matrix Systems	MTX	☒	172	MTX-xxxx	
12	20 ea	3022218	End bracket, material: PA, color: gray	Phoenix Contact	RWE	☒	TB1...TB19;TB.G1	MTX-xxxx	
13	3 ea	3044173	Protective conductor terminal block, number of connections: 2, connection method: Screw connection, Rated cross section: 10 mm², cross section: 0.5 mm² - 16 mm², mounting type: NS 35/7,5, NS 35/15, color: green-yellow	Phoenix Contact	RWE	☒	TB.GND	MTX-xxxx	
14	1 ea	HDL36025	Circuit breaker, PowerPacT H, 25A, 3 pole, 600VAC, 14kA, lugs, thermal magnetic, 80%	Schneider Electric	RWE	☒	CB102	MTX-xxxx	
15	1 ea	S29338	Circuit breaker accessory, PowerPacT H/J, rotary handle, door mounted, Disconnect switch	Schneider Electric	RWE	☒	CB102	MTX-xxxx	

NOTES

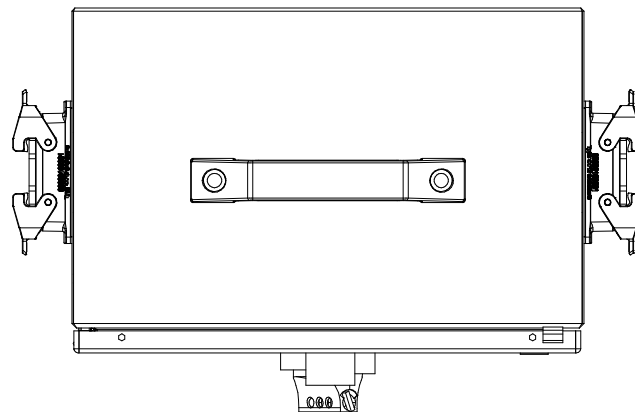
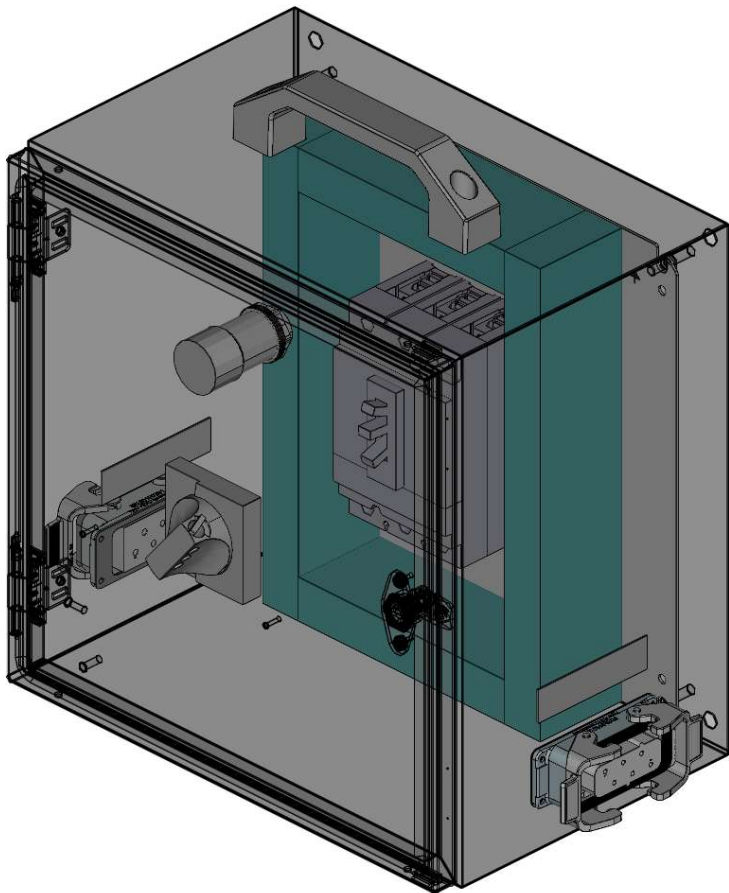
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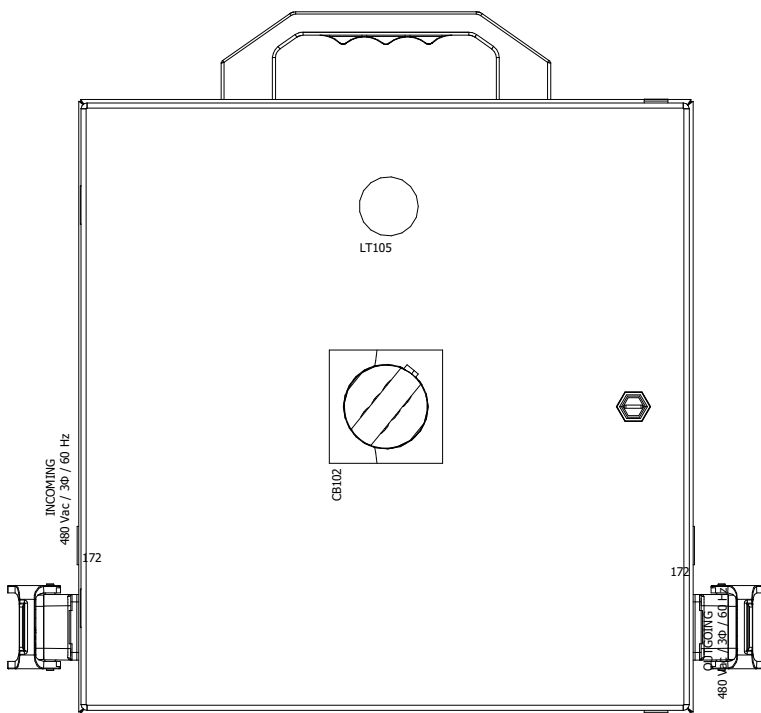


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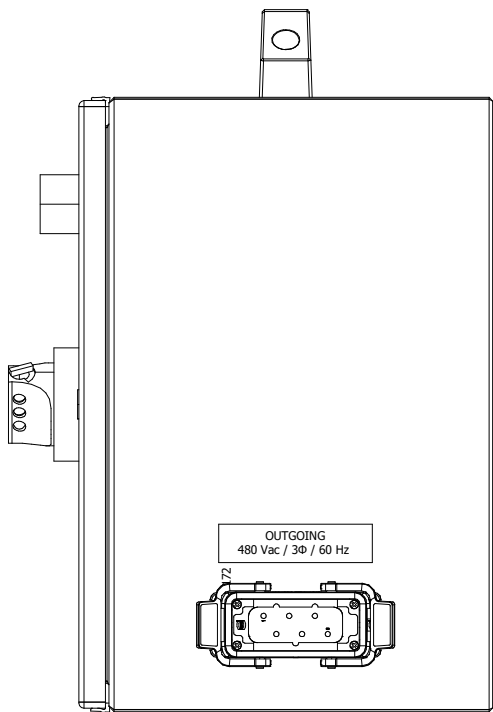
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Top



Front



Right

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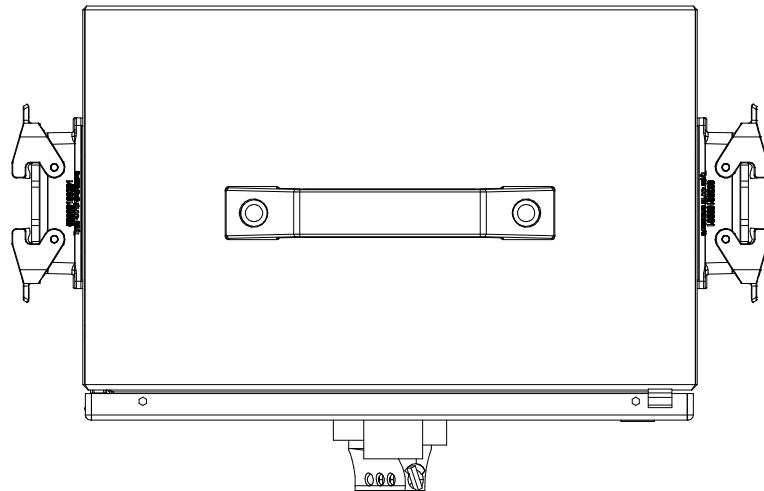
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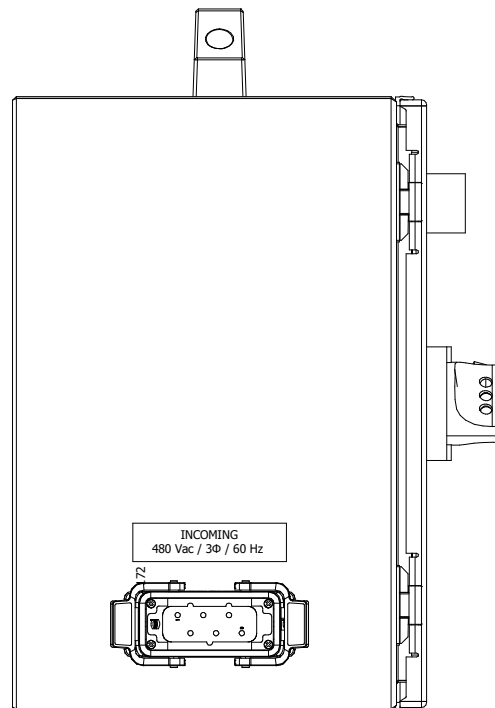
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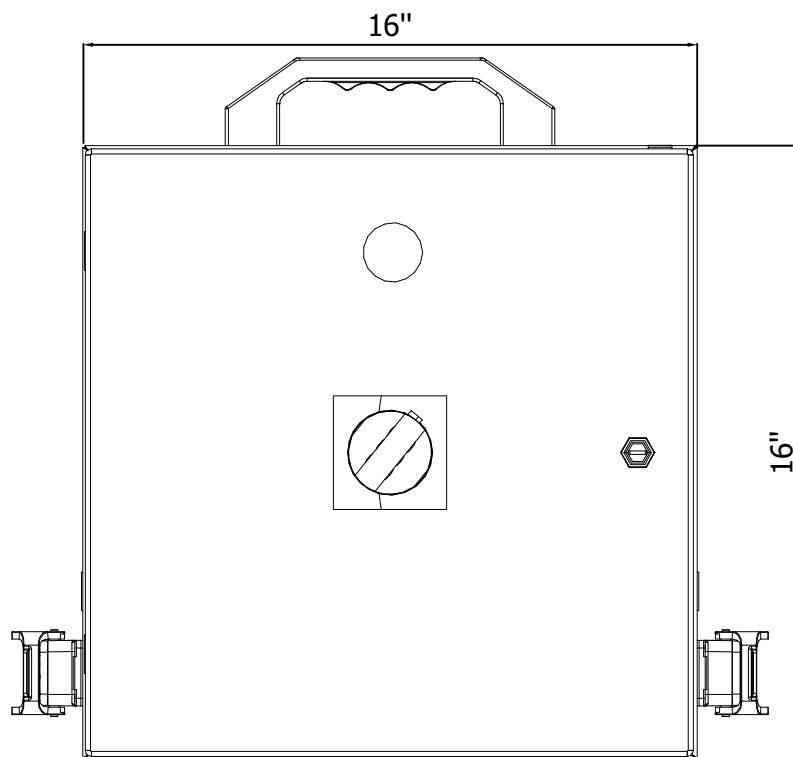
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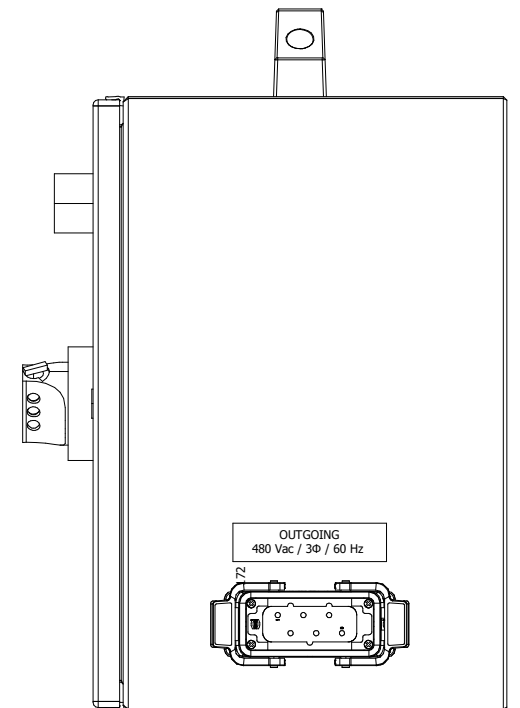
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Left



Front



Right

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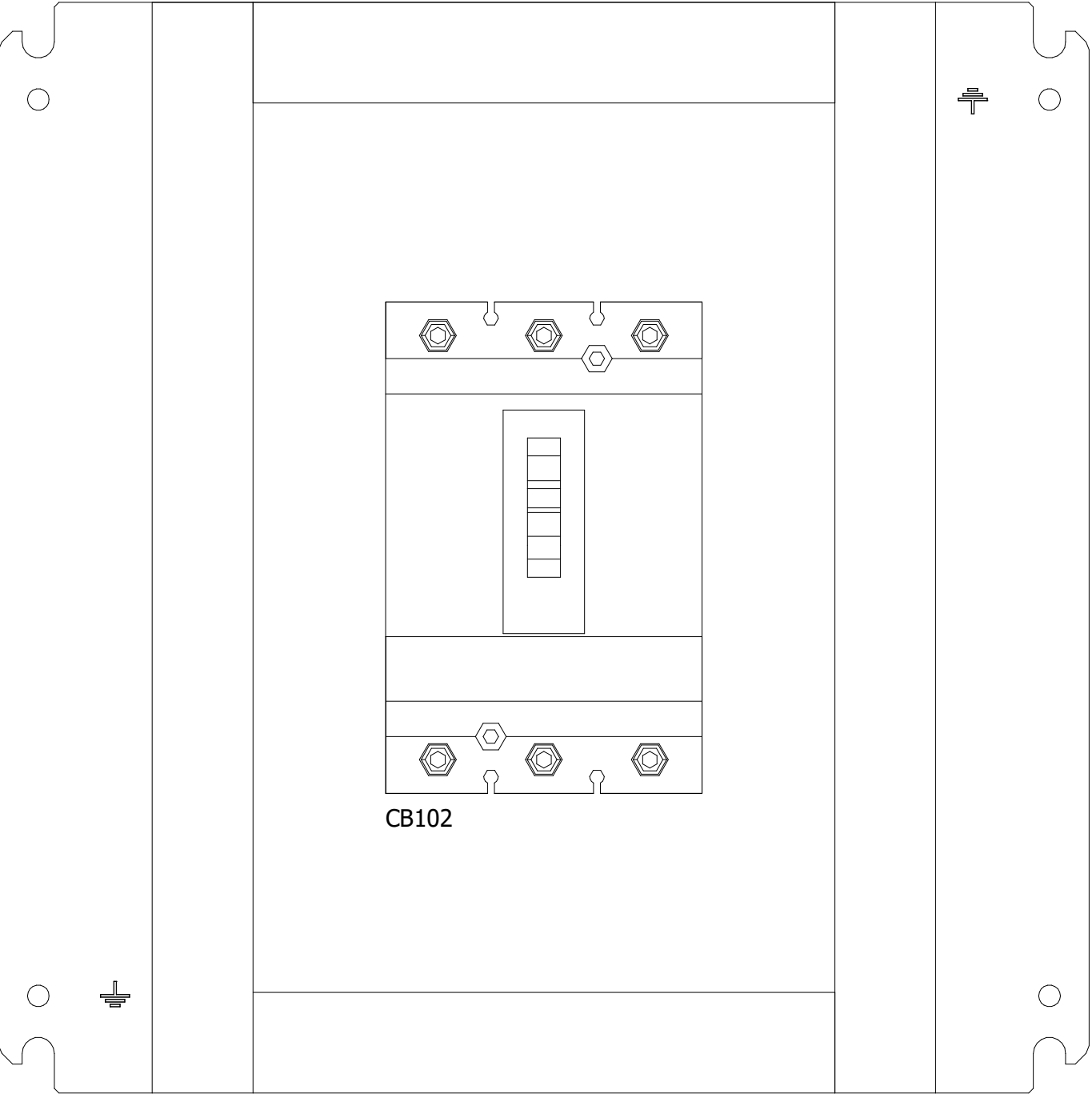
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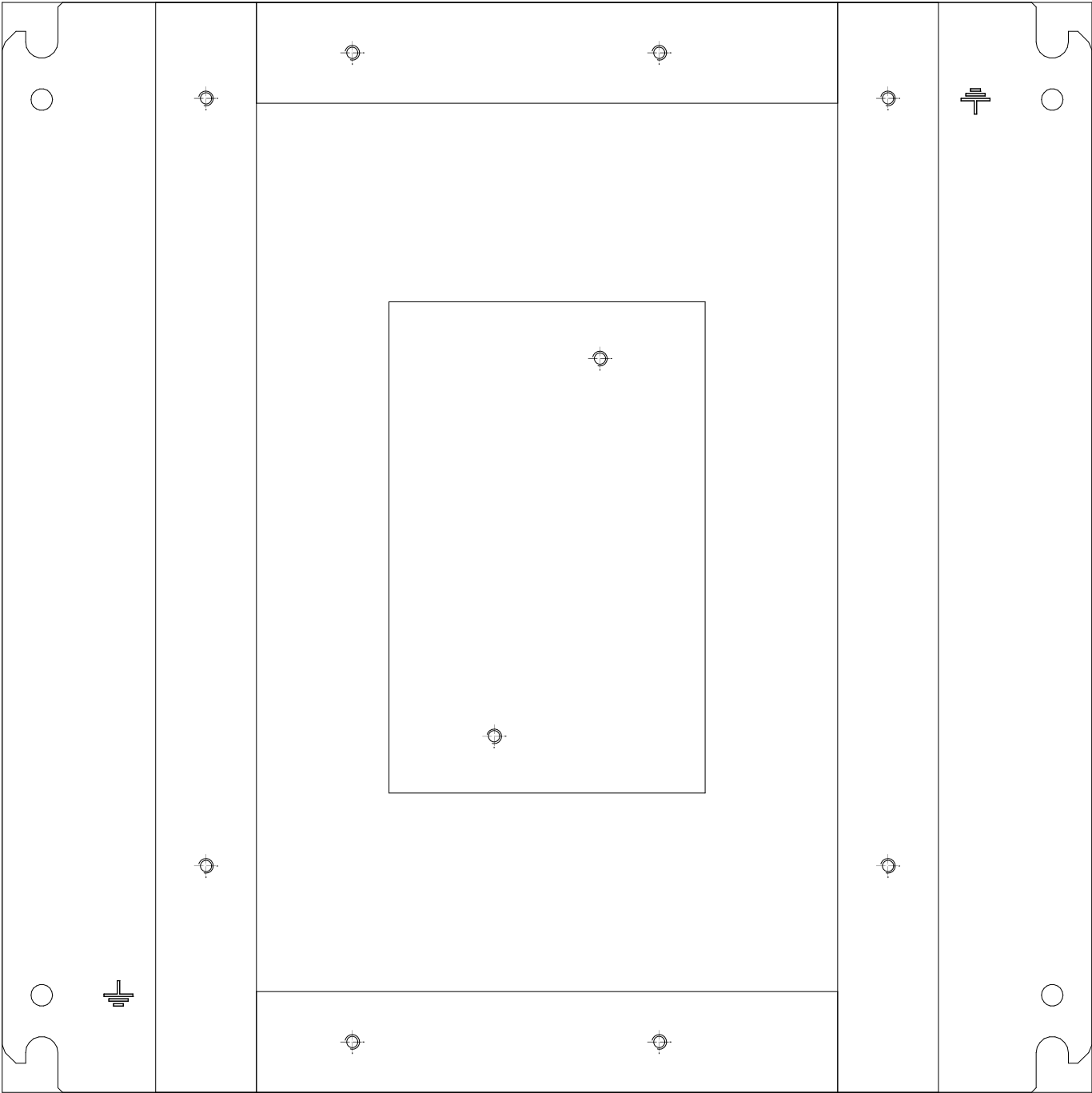
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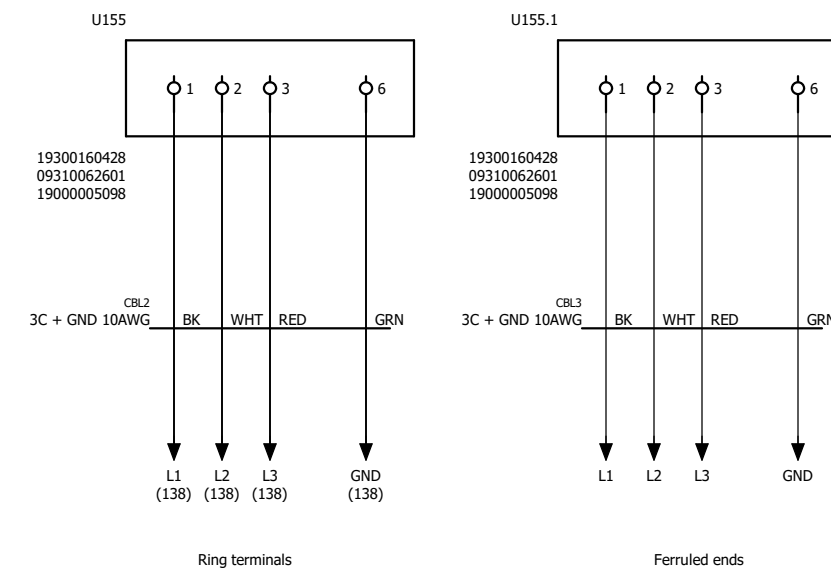
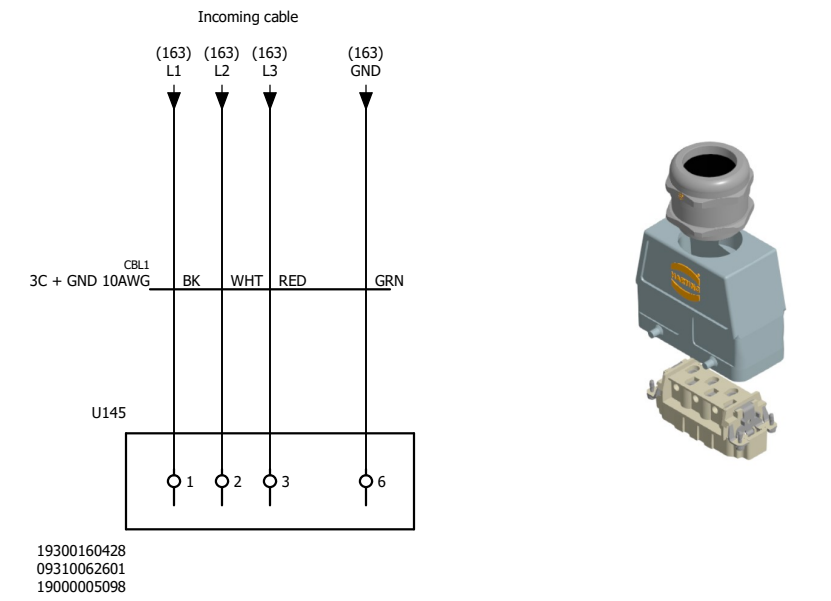
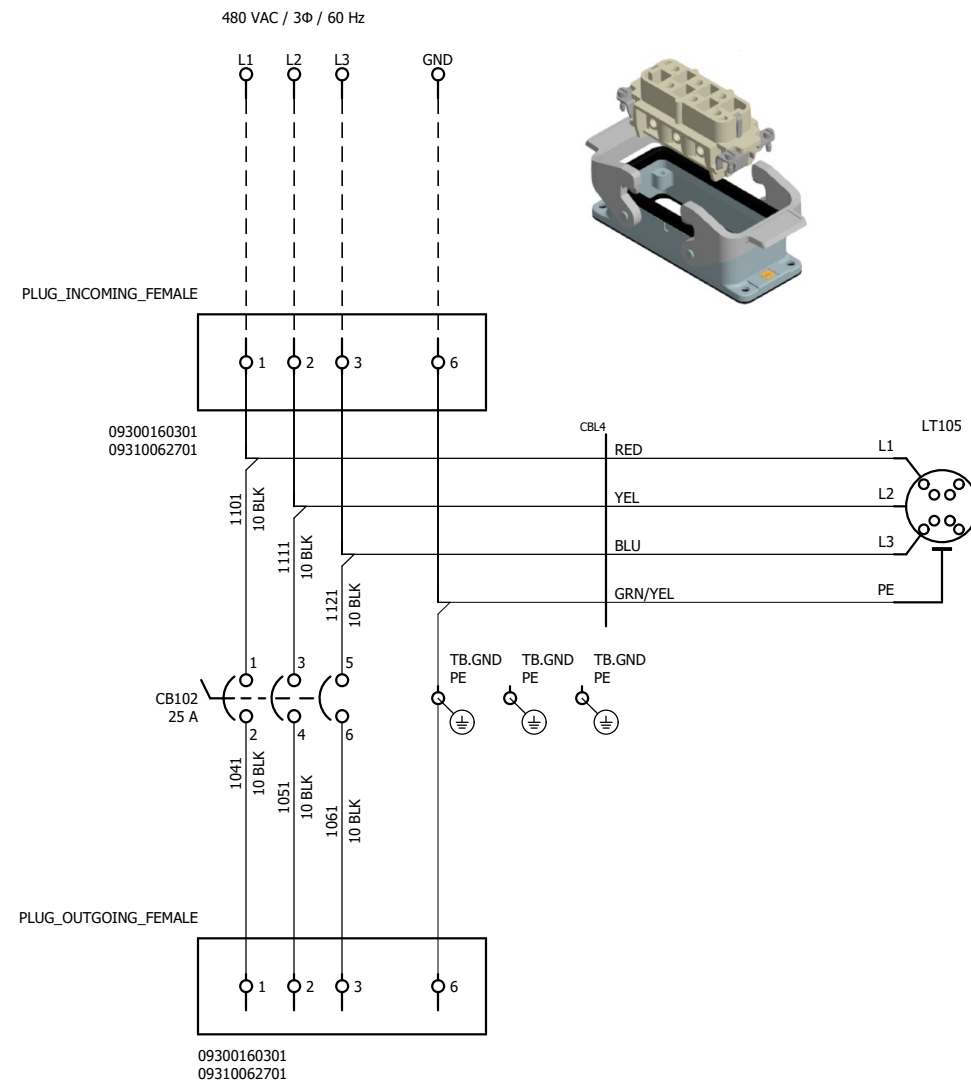


CB102

Internal General Arrangement



Drilling View



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Connection list

F27_002_en_US

Connection	Target 1	Target 2	Color	Cross-section
	-TB.GND:PE:2	-PLUG_OUTGOING_FEMALE	GRN	14
1061	-CB102:6	-PLUG_OUTGOING_FEMALE	BLK	10
1051	-CB102:4	-PLUG_OUTGOING_FEMALE	BLK	10
1041	-CB102:2	-PLUG_OUTGOING_FEMALE	BLK	10

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